

A Long Short-term Memory Autoencoder Approach for EEG Motor Imagery Classification

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Abstract— Motor imagery represents one Brain-Computer Interface (BCI) paradigm that has been utilized in developing applications to assist subjects with motor disability. Such paradigm relies on analyzing brain electroencephalography (EEG) activity to identify the intended movement direction. Existing motor imagery feature extraction techniques are focused on utilizing traditional signal processing and machine learning techniques. Recent advances in the deep learning field has inspired the development of few methods for motor imagery classification that achieved further performance improvement. This paper proposes a deep neural network approach for motor imagery classification using Long Short-Term Memory (LSTM) combined with Autoencoders based on a sequence-to-sequence architecture. The proposed network extracts features from the frequency-domain representation of EEG signals. This network is trained to obtain low-dimensional representation of EEG features that are then fed into multi-layer perceptron of 3 layers for classification. Systematic and extensive examinations have been carried out by applying the approach to public benchmark EEG datasets. The obtained results outperformed classical state-of-the-art methods employing standard frequency-domain features and common spatial patterns, and comparative results to methods such as filter bank common spatial pattern and its variants. Our results indicate the efficacy of the proposed LSTM autoencoder approach in EEG motor imagery classification.

Keywords— Motor imagery, deep learning, long short-term memory, brain computer interface, EEG

I. INTRODUCTION

A Brain-Computer Interface (BCI) system is a recognition and communication system between the human brain and the computer[1, 2]. A BCI system aims at processing brain signals to recognize human intention that is subsequently used to control an external device. BCI systems applications are growing widely from keyboard typing and video gaming to controlling robotic arms and wheelchairs for handicapped patients[3]. BCI systems are based on monitoring and analyzing electroencephalography (EEG) brain activity. EEG activity is a non-invasive signal that can be produced through different signals including P300 evoked potentials, steady-state visual evoked potentials (SSVEP) and motor imagery (MI)[3]. Motor imagery (MI) signals are evoked during the imagination of movement without performing the actual action. Various studies have shown that different signals are observed in the sensorimotor cortex of human brain corresponding to movements in different directions. Thus, machine learning classification is

typically utilized to recognize the EEG signals corresponding to the intended movement.

Classical machine learning approaches have traditionally been used in developing MI BCI systems[4]. However, more recently, deep learning approaches have been utilized in classifying MI EEG signals. For instance, Li et al. proposed denoising autoencoder (DAE) for incomplete EEG signals on spectral power features to classify EEG signals. This approach uses mapping of fine-tuned weights of the DAE to initialize the neural network for motor imagery recognition[5]. DAE is used in decoding incomplete and noisy EEG signals to enhance BCI applications. Yang et al. proposed the use of Convolutional Neural Networks (CNNs) based on augmented Common Spatial Filter (CSP) of EEG signals[6]. In this approach, augmented CSP is used for feature extraction applied to EEG time-domain signals followed by a classification stage that uses CNN. Their reported results indicate better performance than filter-bank common spatial filter (FBCSP)[7].

Other deep learning techniques employed deep belief networks. For instance, An et al. proposed the combination of deep belief network (DBN) and Ada-boost to process EEG time-domain signals[8]. The method uses a combination of trained weak classifiers fed to DBN followed by a classification stage applied to EEG time series. The approach has been tested on recorded data of 4 subjects with 30 trials for right and left motor imagery signals achieving 80.5% accuracy. Na et al. proposed frequential deep belief network (FDBN) based on a restricted Boltzmann machine (RBM) applied to frequency-domain features[9]. The approach uses unsupervised pretrained RBM to feed to DBN training for classification. Reported results indicate a better performance compared to state-of-the-art methods.

In this paper, we introduce a deep learning approach in solving the motor imagery classification problem that adopts Long Short-term Memory (LSTM) neural networks and autoencoders. The sequential and dynamic nature of EEG signals motivate the use of this type of neural networks in such problem. Our results demonstrate performance improvement on benchmark dataset subjects competing with state-of-the-art motor imagery classification approaches.

II. METHODS

A. Datasets and Experiments

In this study, the BCI competition IV dataset 2b is used to verify the performance of the proposed approach[10]. The dataset includes data from 9 subjects with 5 sessions of motor imagery data recorded from 3 EEG channels; namely, C3, Cz and C4. Motor imagery training was based on two main actions; the first is to imagine the corresponding hand movement of presented direction, and the second is trying to move a smiley face shape to the presented direction using motor imagery. The movement of the face on the screen provides feedback to the user.

The dataset of each subject consists of 5 data sessions. The first two sessions are recorded without feedback with 120 trials and the other three sessions with online feedback with 160 trials. Each trial is 9 seconds long with only 4 seconds of motor imagery, indexed from second 3 to second 7. This dataset was recorded with 250 Hz sampling frequency with two classes of motor imageries; left-hand and right-hand movement.

B. Proposed Approach Overview

Our proposed approach is based on using an autoencoder with LSTMs for motor imagery classification. In order to train the LSTM autoencoder, EEG signals are first filtered using bandpass filtering followed by applying Fast Fourier Transform (FFT) to obtain frequency-domain representation. Normalized frequency features are then fed to our network as shown in Fig. 1.

C. Data Pre-processing

It has been demonstrated in numerous studies that human brain activity corresponding to the response of motor imagery can be represented in five frequency bands [1, 3, 11, 12]. Those bands are Alpha, Sigma, low Beta, high Beta and Gamma each has range of (8-13 Hz), (11-15 Hz), (18-23 Hz), (21-26 Hz) and (25-35 Hz), respectively. Therefore, bandpass filters are first applied in our approach to exclude unwanted ranges and noise that take more discriminative bands to simplify feature extraction. A fifth-order Butterworth bandpass filter is applied to the signals recorded on channels C3 and C4 in the range 8 to 35 Hz after filtering trials artifacts [10]. We then apply Fast-Fourier Transform (FFT) with a window size of 1024 [9] which has been reduced to 110 data points per channel by selecting frequencies between 8 and 35 Hz indexed by

$$i = \frac{f_i}{F_s/N} \quad (1)$$

where i is the frequency index, f_i is the frequency to access, F_s is the sampling frequency and N is the number of frequency samples. Combining C3 and C4 channels resulted in 220 data points that are then split into 55 timesteps for training the network.

The Autoencoder task is to minimize such reconstruction error. The encoder stage of the autoencoder takes an input X

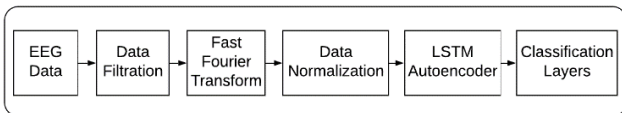


Fig. 1. LSTM autoencoder scheme for motor imagery classification.

D. Approach

1) Autoencoder

An autoencoder consists of an encoder φ and a decoder δ where [13]

$$\varphi: X \rightarrow S, \quad (2)$$

$$\delta: S \rightarrow X \quad (3)$$

where X is the input and S is the encoded representation. The autoencoder attempts to obtain φ and δ by solving [14]

$$\varphi, \delta = \arg \min_{\varphi', \delta'} \|X - (\varphi' \cdot \delta')X\|^2 \quad (4)$$

and maps it into S , $S = \sigma(Wx + b)$, where S is latent representation, W is weight matrix and b is a bias vector. The decoder stage then maps S to $\hat{S}$ with encoder latent states: $\hat{S} = \delta(\hat{W}x + \hat{b})$, where $\hat{W}$ and $\hat{b}$ are for the decoder stage.

2) Long Short-term Memory

Long Short-term Memory (LSTM) has three gates: the forget gate f_t , the update gate i_t and the output gate O_t [15]. The output at time t is h_t , where an LSTM cell state at time t is C_t that encounters previous output h_{t-1} and current input x_t .

LSTM current cell state C_t is computed from previous cell state and the current cell state value [16]

$$C_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c), \quad (5)$$

$$C_t = f_t * C_{t-1} + i_t * C_t \quad (6)$$

where C_t is the computed cell state, W_c is the weight matrix, h_{t-1} is the previous output, x_t is the current input, b_c is the cell bias, f_t is the forget gate state, i_t is the update gate state, C_t is the current cell state and C_{t-1} is the previous cell state.

The forget gate passes most relevant information from the previous cell state to the current cell state. The update gate updates the current cell state with value of $\hat{C}_t$. Each of the update and the forget gates have a value between 0 and 1. Therefore, the LSTM output O_t is a function of the cell state that has been updated through time.

3) LSTM Autoencoder

The LSTM autoencoder has the ability to encode multidimensional features into lower dimensional representations [17] that has been shown useful in pattern identification and also anomaly detection [18]. Thus, it helps in learning from high dimensional data with low number of records.

Our proposed approach is inspired by sequence-to-sequence tasks that are typically used in machine translation tasks, text generation and time series prediction [1, 2]. LSTM Autoencoder (LSTM-AE) is an unsupervised learning technique that feeds the sequence of data to the network that transforms it into compressed representation combined with LSTM as hidden nodes [3, 15]. The output is then provided to the decoder with the same input using encoder latent states to build up a new representation. This representation describes the whole sequence as a fixed-size embedding vector in which the encoder is fed with $x(t, \dots, t + N - 1)$ sequence input while the decoder is provided with a shifted version with one timestep that takes the form $x(t + 1, \dots, t + N)$.

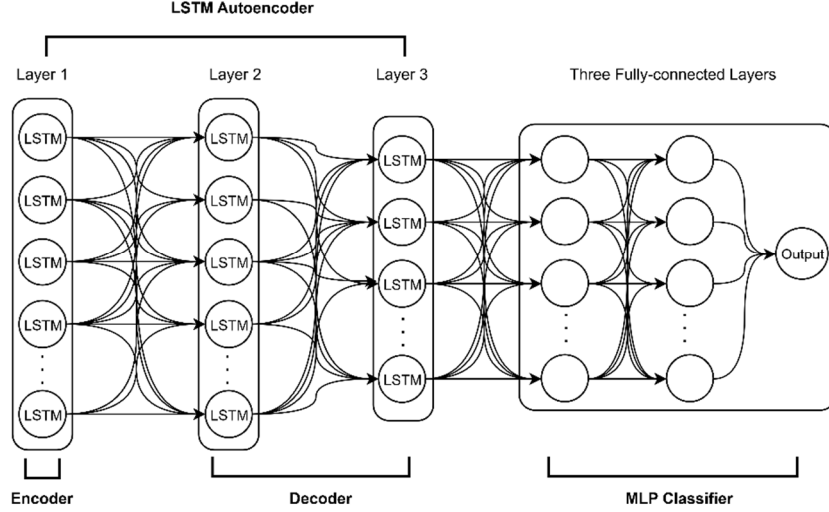


Fig. 2. Proposed LSTM autoencoder approach for motor imagery classification with 3 LSTM autoencoder layers followed by three fully connected layers. Layers 1, 2 and 3 in the figure correspond to the hidden layers of the trained autoencoders.

In our analysis, we used EEG session 1 and session 2 for training combined with splitting session 3 into train and validation sets with 80% and 20% of the data, respectively. We used data augmentation to improve the learning process while applying L2 normalization and robust scaling where L2 normalization is defined by

$$\|\hat{x}\| = \frac{x_i}{\sqrt{\sum_i^n (x_i)^2}} \quad (7)$$

and robust scaling is defined by

$$\|\hat{x}\| = \frac{x_i - Q_1(x)}{Q_3(x) - Q_1(x)} \quad (8)$$

where x is the input data, $\hat{x}$ is normalized data, $Q_1(x)$ and $Q_3(x)$ are first and third quartiles of calculated interquartile range (IQR) of input data, respectively.

The proposed LSTM autoencoder approach consists of two hidden layers with sizes 16 and 8, respectively, using LSTM as hidden nodes. We then add a Multi-Layer Perceptron (MLP) classifier that includes 3 fully connected layers with 8, 8 and 1 nodes, respectively, as shown in Fig. 2. ReLU and leaky-ReLU activation functions have been used that take the form [19]

$$\text{ReLU:} \quad R(x) = \begin{cases} x & x > 0 \\ 0 & x \leq 0 \end{cases} \quad (9)$$

$$\text{Leaky-ReLU:} \quad R(x) = \begin{cases} x & x > 0 \\ \alpha x & x \leq 0 \end{cases} \quad (10)$$

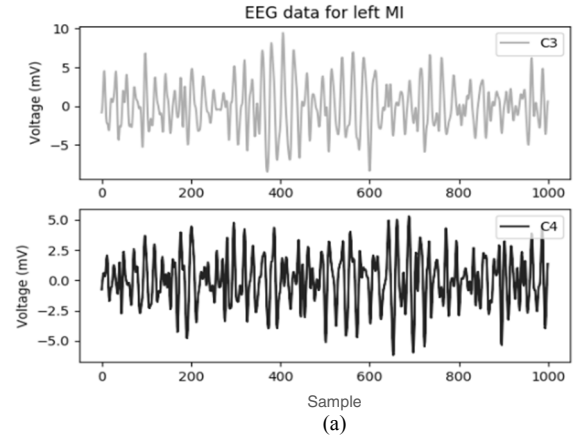
where x is the input data, α is slope coefficient of negative values and $R(x)$ is the rectified output.

To train the proposed LSTM-AE, Keras framework has been used on a machine with Tesla-k80 GPU. The number of model trainable parameters is 6,969 that required 572 MB of memory. Extensive experiments have been conducted on the BCI competition IV dataset 2b for training LSTM-AE using different sets of hyper-parameters that are combination of “logcosh” loss function, “adamax” optimizer, LSTM dropout of 0.1, MLP dropout of (0.01, 0.05 or 0.3) and (0.05 or 0.1), for connections between MLP layers 1 and 2, and MLP layers 2 and 3, respectively, learning rate (0.0015 or

0.005), epochs 550 and data augmentation. In our approach, optimal hyper-parameters for each subject are obtained independently using permutations of the aforementioned parameters.

III. RESULTS

The main objective of this paper is to classify recorded EEG signals to either representing left or right motor imagery (MI) activities. Fig. 3 demonstrates 1000 samples (corresponding to 4 seconds) of the recorded data where the motor imagery tasks was being performed between seconds 3 and 7. The figure illustrates the dynamic nature of the recorded EEG in both cases with no distinctive pattern for



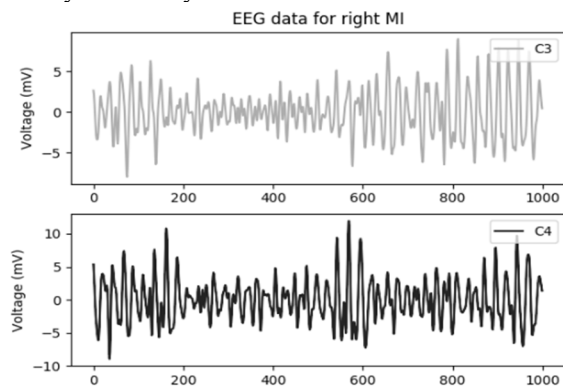


Fig. 3. EEG time-domain signals recorded on channels 3 and 4 for (a) left and (b) right motor imagery tasks.

TABLE I. LSTM-AE SESSIONS 4 AND 5 CLASSIFICATION ACCURACY.

Subject	1	2	3	4	5	6	7	8	9	Avg.
Session 4	85	63	61	98	88	88	79	90	88	82
Session 5	65	66	57	94	98	91	80	89	79	80
Avg.	75	65	59	96	93	90	80	90	84	81

TABLE II. COMPARISON BETWEEN LSTM-AE AND STATE-OF-THE-ART METHODS.

Subject	Chin	Gan	Coyle	Bi-spectrum	CSP	FBCSP MIBIF	FBCSP MIRSIR	FDBN	LSTM-AE
1	70	71	60	77	66	68	70	81	75
2	61	61	56	65	62	59	61	65	65
3	61	57	56	61	57	59	61	66	59
4	98	97	89	97	97	98	98	98	96
5	93	86	79	82	77	93	93	93	93
6	81	81	75	85	75	80	81	88	90
7	78	81	69	75	77	78	78	82	80
8	93	92	93	91	93	93	93	94	92
9	87	89	81	87	83	88	87	90	84
Avg.	80	79	73	80	76	80	80	84	81

each task. This justifies the need for using sophisticated deep learning techniques to capture the underlying complexity of EEG data.

A. Performance of the LSTM-AE Approach

To evaluate the performance of LSTM-AE for motor imagery classification, we used the BCI competition IV dataset using systematic experiments with sessions 4 and 5 used as the test dataset [10]. In this analysis, EEG signals recorded in sessions 1 to 3 of each subject were used to train a subject-dependent LSTM-AE that was subsequently used to classify the signals of sessions 4 and 5. Table I shows the classification accuracy results obtained for each of the 9 subjects for each of the two sessions. Results demonstrate an average classification accuracy across all subjects of 81%, with the highest classification accuracy obtained for Subject 4 (96%).

B. Comparison of LSTM-AE with State-of-the-art Methods

We compared the performance of the LSTM-AE approach to methods that were applied to the BCI

competition IV dataset in the literature including the three top accuracies obtained by researchers who participated in the competition (Chin, Gan and Coyle). We also compared LSTM-AE to other state-of-the-art methods as shown in Table II including Bispectrum-based approach[20], Common Spatial Filter (CSP)[21], Filter Bank Common Spatial Pattern (FBCSP) with Mutual Information-based Best Individual Feature (MIBIF)[7], FBCSP with Mutual Information-based Rough Set Reduction (MIRSIR)[7] and Frequential Deep Belief Network (FDBN) based on a Restricted Boltzmann Machine (RBM)[9].

Comparing the performance of the LSTM-AE approach to the three top-ranked methods participated in the competition, our results indicate that the LSTM-AE approach outperforms all three approaches, where the average accuracy of the methods proposed by Chin, Gan and Coyle are 80%, 79% and 73%, respectively. The proposed LSTM-AE approach outperformed all 3 approaches on average and specifically in a total of 4 subjects. It should be noted that three approaches relied primarily on classical machine learning classifiers. This indicates how deep learning

approaches, especially LSTM-based ones, are more suited for this type of data.

Comparing the LSTM-AE to the Bi-spectrum, CSP, FBCSP and MIRS approaches, LSTM-AE achieved outperforming or equal performance in 4 subjects. Our results also indicate that the accuracy obtained for Subject 6 has been improved to 90% that possessed the best performance, which is higher than the best performing approach (FDBN), that has much higher computational complexity, and with similar accuracy achieved for subjects 2 and 5.

C. Performance of the LSTM-AE Approach for Different Dropout Ratios

We next examined the effect of varying the dropout ratio value of the connections between MLP layers 1 and 2 on the performance of LSTM-AE. For this analysis, the dropout ratio for the connections between MLP layers 2 and 3 was fixed at 0.05. We examined three dropout ratios, namely 0.05, 0.1 and 0.3. Fig. 4 illustrates the mean accuracy of sessions 4 and 5 for each subject for each of the three examined dropout ratios. Averaging across subjects, the mean accuracy for dropout ratios 0.05, 0.1 and 0.3 are 80%, 79% and 78%, respectively. This indicates a slight enhancement in performance for lower dropout ratios. It should be noted that the results reported

TABLE III. PERFORMANCE OF LSTM-AE USING SUBJECT 9 DATASET AS TRAINING DATASET.

Subject	1	2	3	4	5	6	7	8	9	Avg.
Session 4	66	58	49	94	63	84	63	66	88	70
Session 5	69	48	58	87	61	79	64	73	79	69
Avg.	68	53	53	90	62	81	63	69	84	69

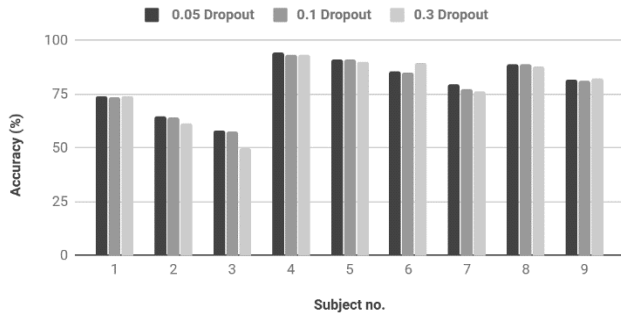


Fig. 4. Performance of the LSTM-AE approach for different dropout ratios for the connections between MLP layers 1 and 2.

previously in Table I and Table II were achieved using the best dropout ratio tuned for each subject independently. Finally, varying the dropout ratio for the connections between MLP layers 2 and 3 did not have a significant impact on the achieved accuracies.

D. Across Subjects Performance of LSTM-AE

To examine the ability of the proposed LSTM-AE to generalize across subjects, we examined the performance of the proposed approach when the training dataset of one subject only is used to train the LSTM-AE, and subsequently use this LSTM-AE to classify the test datasets of all other subjects. Table III demonstrates the average accuracy obtained for sessions 4 and 5 of each subject while training LSTM-AE with common architecture configuration across all subjects using the data of Subject 9 (chosen as one of the average-performing subjects). An average accuracy of 69% is achieved in this case compared to 81% accuracy obtained using subject-dependent training. This indicates the ability of the LSTM-AE to generalize across subjects, albeit with lower accuracy compared to subject-dependent training.

IV. CONCLUSION

Motor imagery represents one of the major paradigms of BCI systems. In this paper, we introduced a deep learning scheme based on LSTM Autoencoder for motor imagery

classification. Our results indicate that the proposed approach achieves an average accuracy of 81% which can compete with and outperform state-of-the-art methods in some cases. Moreover, the proposed approach can handle subject-to-subject variability indicating the ability of the approach to generalize which is expected to improve by training with more data of EEG motor imagery. Further enhancements could be achieved using more advanced signal pre-processing techniques as well as increasing the size of the dataset.

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